Electric Potential:

→ Potential difference is defined as work done per unit charge in bringing a positive test charge from pt. b to a in electrostatics. [Note: charge is moved slowly, with no change in KE].



$$V(b) - V(a) = -\int_a^b \vec{E} \cdot d\vec{l}$$

→ Now potential at a point is defined as moving a positive test charge from ∞ to that pt.

$$\therefore V(r) = -\int_{\infty}^{r} \vec{E} \cdot d\vec{l}$$

∵ External agent has to do the work against the field we take it as −ve sign [other way to look at is that the $-q_0E$ is the external force opposite to field force]

This is from the fact that

$$V_A = \frac{W_{ext}\big|_{\infty \to A}}{q_0} \quad \& \quad dW = -\vec{F} \cdot d\vec{r}$$

↙ use

$$\vec{F} = q_0 E$$

$$dW = -q_0 \vec{E} \cdot d\vec{r} \quad \} \text{ resultant.}$$

$$\therefore V_A = \int \frac{-q_0 \vec{E} \cdot dr}{q_0} = \int -\vec{E} \cdot d\vec{r}$$

Note: $d\vec{r}$ & $d\vec{l}$ are line elements in this case.

If $\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$

& $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

$$V = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{r} E_x\,dx + E_y\,dy + E_z\,dz$$

$$\therefore E_x = -\frac{\partial V}{\partial x}\ ;\ E_y = -\frac{\partial V}{\partial y}\ ;\ E_z = -\frac{\partial V}{\partial z}$$

$$\therefore \vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$$

$$\Rightarrow \boxed{\vec{E} = -\nabla V}$$

↳ gradient of the potential gives the Electric field.

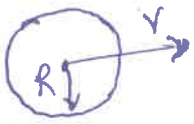
→ For a point charge:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \quad \text{or} \quad \frac{kq}{r^2}\hat{r} \quad \text{where } k = \frac{1}{4\pi\varepsilon_0}$$

$$V = -\int_{\infty}^{r} \vec{E}\cdot dr = -\int_{\infty}^{r} \frac{kq}{r^2}\hat{r}\,dr = \frac{kq}{r}$$

→ for a hollow sphere

a) Outside:



$$\vec{E}_{outside} = \frac{Kq}{r^2}\hat{r}$$

$$\therefore V = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{r} \frac{Kq}{r^2}\hat{r}\cdot(dr)\hat{r}$$

$$\boxed{V = \frac{Kq}{r}} \text{ or } \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

b) Inside:



We now seperate the integral

$$V = -\int_{\infty}^{r} \vec{E}\cdot dr \text{ as } -\int_{\infty}^{R} \vec{E}\cdot d\vec{r} - \int_{R}^{r} \vec{E}\, d\vec{r}$$

∵ the field is different at surface & inside the hollow sphere, we seperate the integrals as from ∞ to R & from R to r.

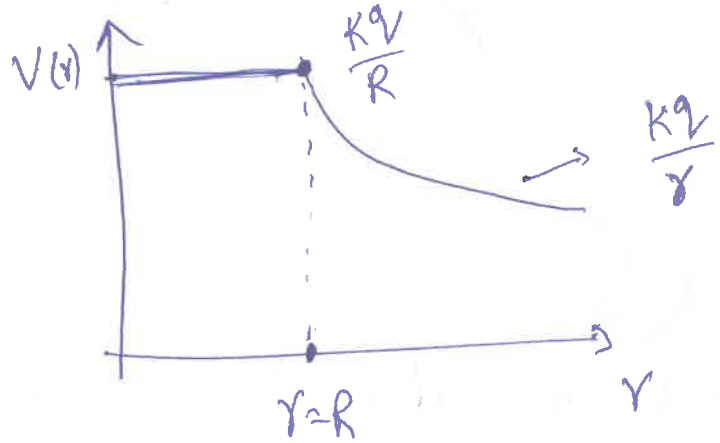
$$\therefore V = -\int_{\infty}^{R} \vec{E}\cdot d\vec{r} - \int_{R}^{r} \vec{E}\cdot d\vec{r}$$

$\vec{E}$ inside hollow sphere is zero $\therefore \vec{E}\cdot d\vec{r} = 0$ in this case

$$\therefore V = -\int_{\infty}^{R} \frac{kq}{r^2}\hat{r}\cdot dr\,\hat{r}$$

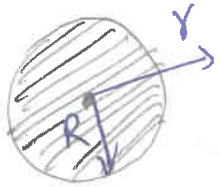
$$= \frac{kq}{R} \text{ or } \frac{1}{4\pi\varepsilon_0}\frac{q}{R}$$

$\therefore$



→ For a Solid Sphere:

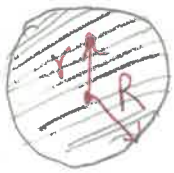
a) Outside



$$\vec{E}_{outside} = \frac{kq}{r^2}\hat{r}$$

$$\therefore V_{outside} = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = \frac{kq}{r} \text{ or } \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

b) Inside :



Once again we split the integral from ∞ to R & R to r.

$$\therefore V = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{R} \frac{kq}{r^2}\hat{r}\cdot dr\hat{r} - \int_{R}^{r} \underbrace{\frac{kq}{R^3} r\hat{r}\cdot dr\hat{r}}_{\text{field inside solid sphere.}}$$

$$= \frac{kq}{R} - \frac{kq}{R^3}\int_{R}^{r} r\,dr$$

$$= \frac{kq}{R} - \frac{kq}{R^3}\left[\frac{r^2}{2}\Big|_{R}^{r}\right]$$

$$= \frac{kq}{R} - \frac{kq\,r^2}{2R^3} + \frac{kq\,R^2}{2R^3}$$

$$\boxed{V = \frac{kq}{2R}\left[3 - \frac{r^2}{R^2}\right]}$$

→ Potential inside a uniformly charged solid sphere.

at $r = 0$ (Center of the sphere) $V = \frac{3kq}{2R} = 1.5\,V_{surface}$

Potential energy

Potential energy of a particle is defined as the amount of work done in moving the particle against a field from ∞ to that pt.

$$\therefore U(r) = W_{done} \propto V(r)$$

$$U(r) = -\int_{\infty}^{r} \vec{F} \cdot d\vec{r}$$

$$\vec{F} = -\nabla U$$

In electro statics

$$\vec{F} = K \frac{q_1 q_2}{r^2} \hat{r}$$ { force between 2 charges q_1 & q_2 seperated by r.

$$\vec{F} = -\nabla U$$

In spherical coordinates

$$-\frac{dU}{dr}\hat{r} = \frac{Kq_1q_2}{r^2}\hat{r}$$

$$\int_0^U dU = \int_{\infty}^{r} -\frac{Kq_1q_2}{r^2} dr \Rightarrow U = \frac{Kq_1q_2}{r}$$